The Islamia University
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Image Classification using Convolutional Neural Network (CNN) on CIFAR-10 Dataset

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Computer Vision is one of the most important fields of Artificial Intelligence that enables machines to understand and analyze visual data. Image classification is a fundamental computer vision task where a model predicts the category of an input image. In this project, a Convolutional Neural Network (CNN) was implemented for image classification using the CIFAR-10 dataset. The model was trained using PyTorch and evaluated using various performance metrics including Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. Data preprocessing and augmentation techniques were applied to improve model generalization. The final model achieved an overall classification accuracy of 77.21%, demonstrating the effectiveness of CNNs for image classification tasks.

1. Introduction

Computer Vision focuses on enabling computers to extract meaningful information from images and videos. One of the most common applications of computer vision is image classification, where an image is assigned to a predefined category.

Traditional machine learning methods required manual feature extraction, which was often time-consuming and less effective. Convolutional Neural Networks (CNNs) automatically learn important image features such as edges, textures, shapes, and object patterns directly from raw images.

The objective of this project is to develop a CNN-based image classification system capable of classifying images from the CIFAR-10 dataset into ten distinct categories.

2. Problem Statement

The goal of this project is to build and train a Convolutional Neural Network capable of accurately classifying images from the CIFAR-10 dataset into one of ten predefined classes. The project aims to evaluate CNN performance using standard classification metrics and visualize the training and evaluation results.

3. Dataset Description

The CIFAR-10 dataset is a widely used benchmark dataset for image classification tasks.

Dataset Characteristics:

- Total Images: 60,000
- Training Images: 50,000
- Testing Images: 10,000
- Number of Classes: 10
- Image Size: 32 × 32 pixels
- Color Channels: RGB

Classes included in the dataset:

1. Airplane
2. Automobile
3. Bird
4. Cat
5. Deer
6. Dog
7. Frog
8. Horse
9. Ship
10. Truck

The dataset provides a balanced distribution of images across all classes.

4. Data Preprocessing

Before training the model, several preprocessing steps were applied:

4.1 Image Normalization

Images were normalized to ensure stable and efficient training.

4.2 Data Augmentation

To improve model generalization and reduce overfitting, the following augmentation techniques were applied:

- Random Horizontal Flip
- Random Rotation
- Tensor Conversion

Data augmentation artificially increases data diversity and helps the model learn robust image features.

5. Methodology

The project follows the standard deep learning workflow:

1. Dataset Loading
2. Data Preprocessing
3. CNN Model Construction
4. Model Training
5. Model Evaluation
6. Results Analysis

The implementation was developed using Python and PyTorch.

6. CNN Architecture

The Convolutional Neural Network architecture consists of multiple layers designed for feature extraction and classification.

Layer Structure

1. Convolution Layer (32 Filters)
2. ReLU Activation Function
3. Max Pooling Layer
4. Convolution Layer (64 Filters)
5. ReLU Activation Function
6. Max Pooling Layer
7. Fully Connected Layer (512 Neurons)
8. Dropout Layer
9. Output Layer (10 Classes)

The convolution layers extract important image features, while the fully connected layers perform classification.

7. Training Procedure

The model was trained using supervised learning.

Training Configuration:

- Optimizer: Adam
- Learning Rate: 0.001
- Batch Size: 64
- Number of Epochs: 20

- Loss Function: CrossEntropyLoss

The Adam optimizer was selected because of its efficiency and fast convergence properties.

8. Experimental Setup

Hardware Environment

- CPU-Based Training Environment

Software Environment

- Python
- PyTorch
- Torchvision
- NumPy
- Matplotlib
- Scikit-learn

9. Performance Evaluation Metrics

The model was evaluated using the following metrics:

Accuracy

Accuracy measures the percentage of correctly classified images.

Precision

Precision measures the proportion of correct positive predictions.

Recall

Recall measures the model's ability to identify actual positive samples.

F1-Score

F1-Score combines Precision and Recall into a single performance metric.

Confusion Matrix

A confusion matrix provides a detailed analysis of prediction performance for each class.

10. Results

The trained CNN achieved the following performance:

Metric	Value
Accuracy	77.21%
Precision	77%
Recall	77%
F1-Score	77%

These results indicate that the CNN successfully learned meaningful image features and achieved good classification performance on the CIFAR-10 dataset.

11. Results Analysis

The confusion matrix revealed that classes such as Automobile, Ship, Truck, and Frog were classified with high accuracy.

However, Cat and Dog classes showed comparatively lower performance. This is expected because:

- Both classes share similar visual characteristics.
- CIFAR-10 images are only 32 × 32 pixels.
- Low image resolution makes feature discrimination more challenging.

The training loss curve demonstrated a steady decrease in loss throughout training, indicating successful learning and convergence of the model.

Figure 1: Training Loss Curve

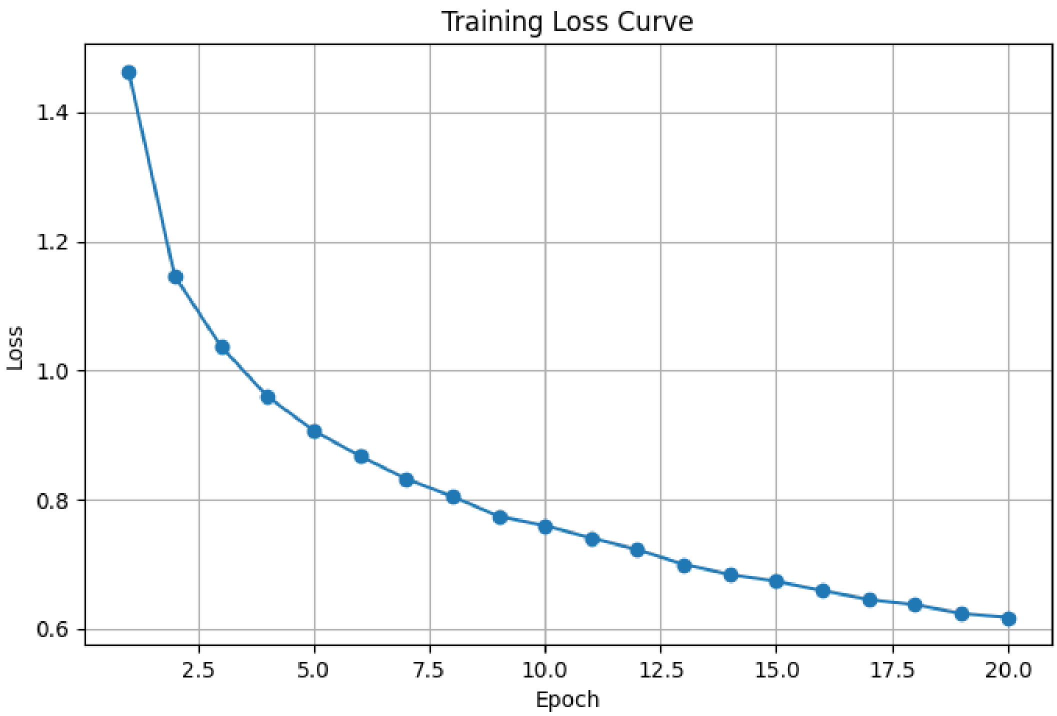

Figure 2: Confusion Matrix

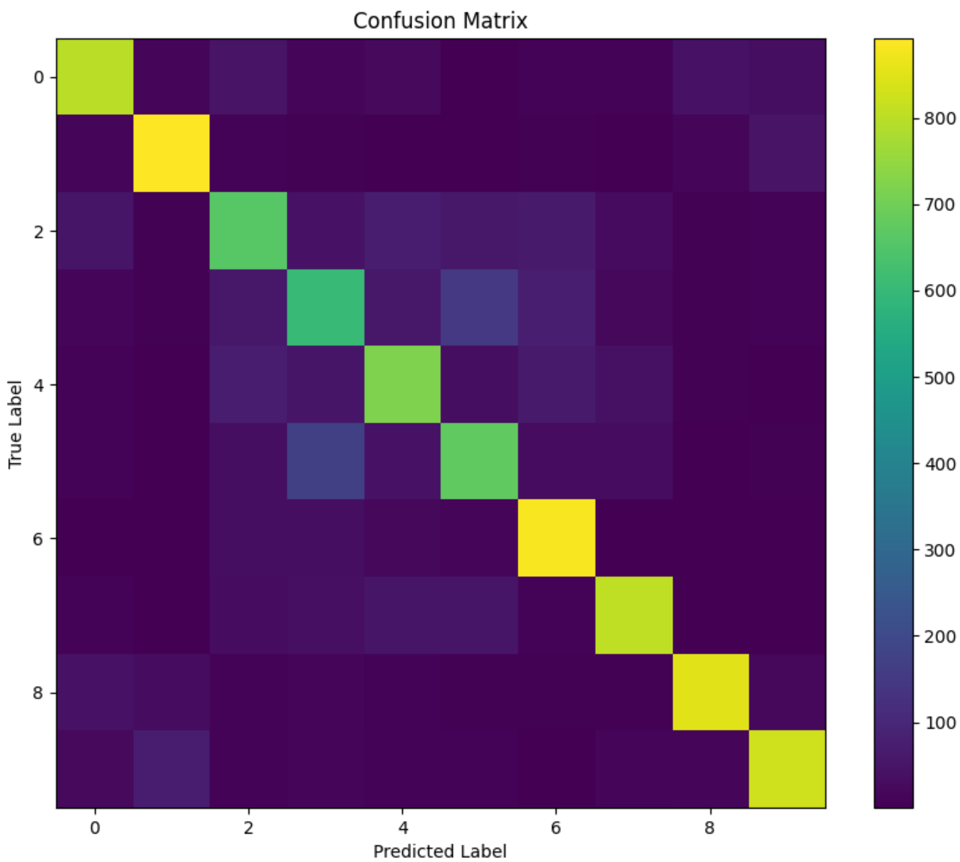

Figure 3: Sample Cat and Dog Images

12. Conclusion

In this project, a Convolutional Neural Network was successfully developed and trained for image classification using the CIFAR-10 dataset. Data preprocessing and augmentation techniques improved the model's performance and generalization capability. The final model achieved an overall accuracy of 77.21%.

The results demonstrate that CNNs are highly effective for image classification tasks and can automatically learn useful visual features without manual feature engineering.

13. Future Work

Future improvements may include:

- Using deeper CNN architectures
- Applying Transfer Learning
- Implementing Vision Transformers (ViT)
- Hyperparameter optimization

- GPU-based training
- Testing on larger image datasets

14. References

1. PyTorch Official Documentation
2. Torchvision Documentation
3. CIFAR-10 Dataset Documentation
4. Goodfellow, I., Bengio, Y., & Courville, A. Deep Learning.
5. Bishop, C. Pattern Recognition and Machine Learning.